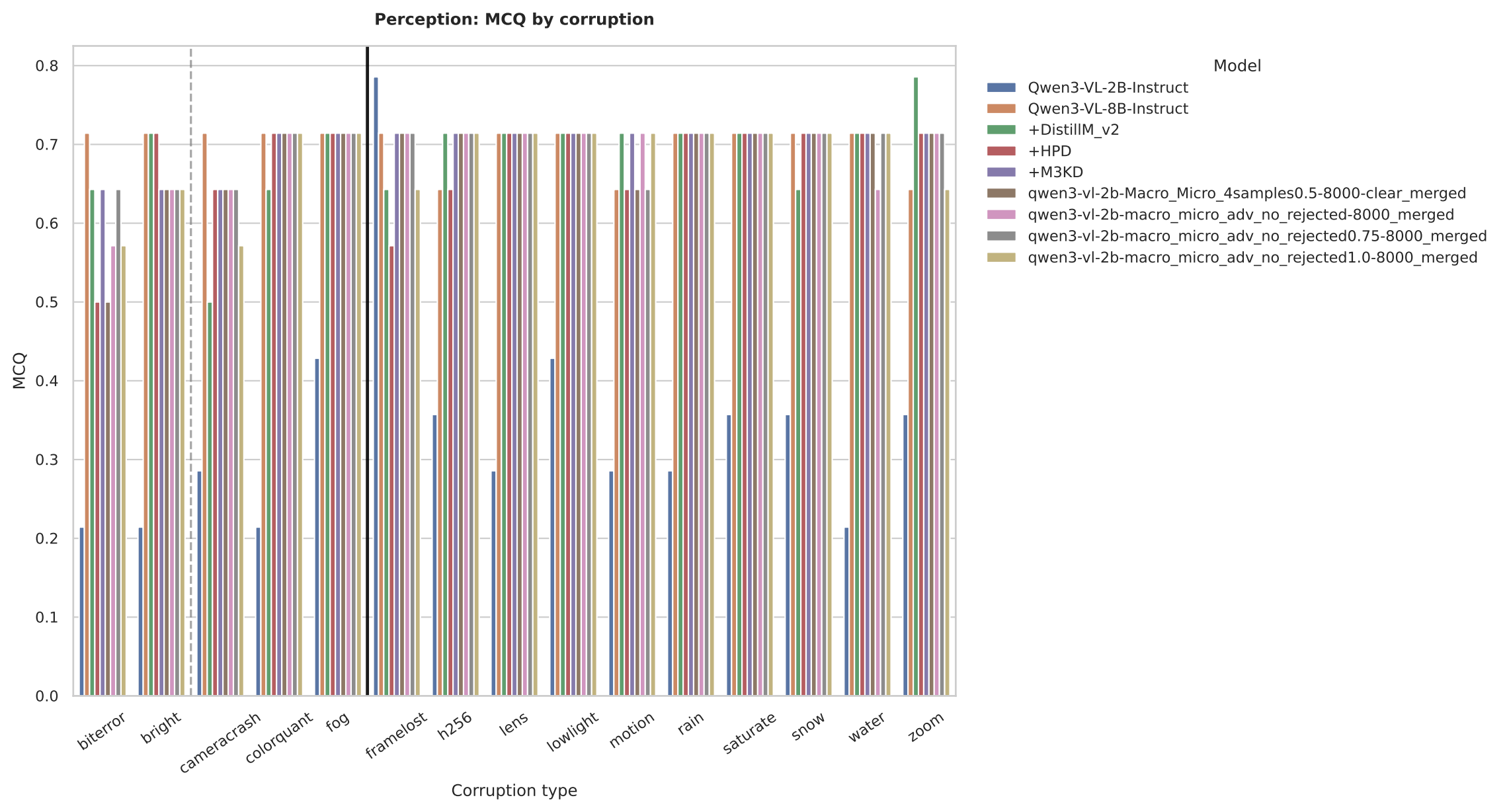
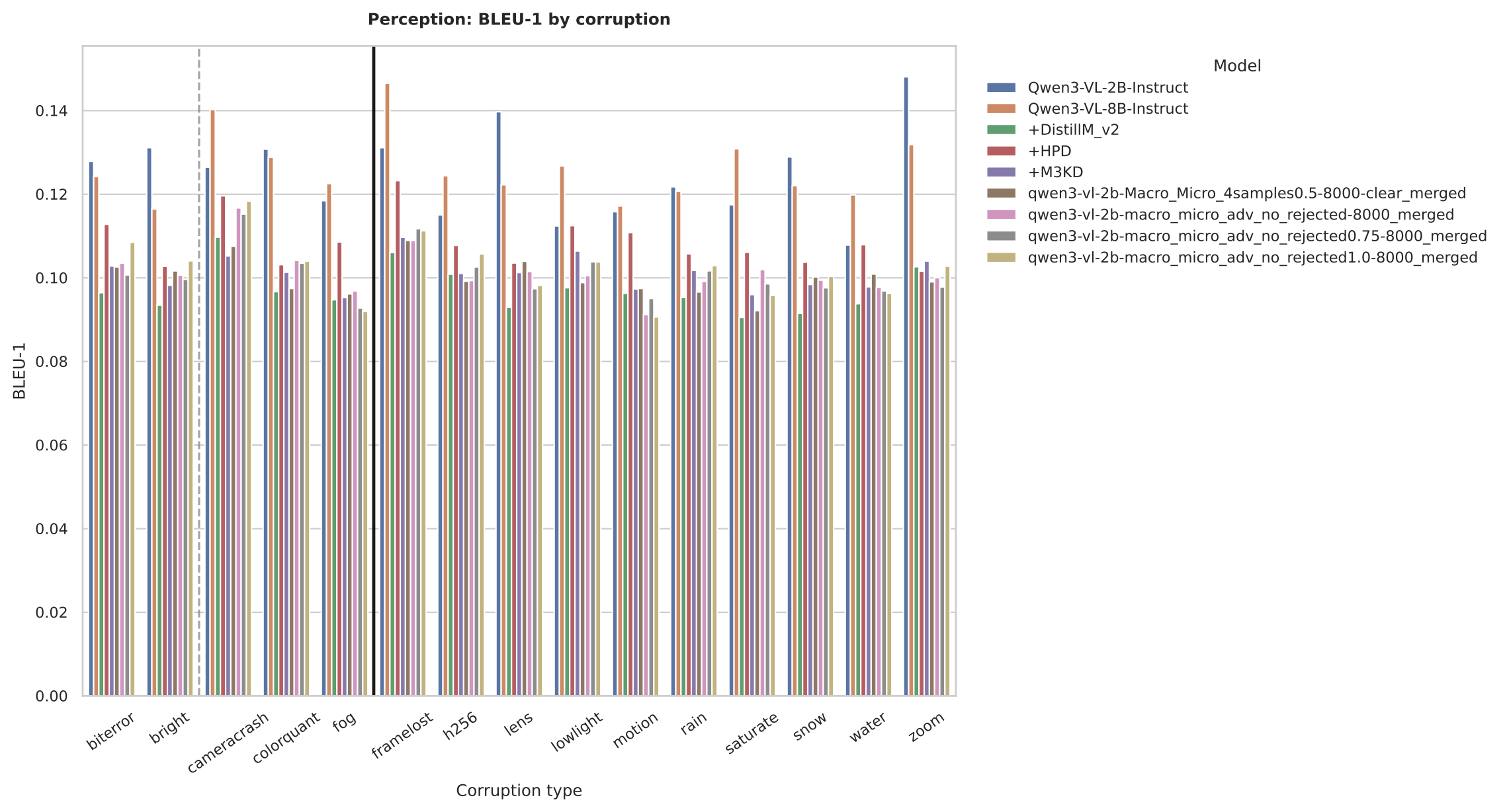
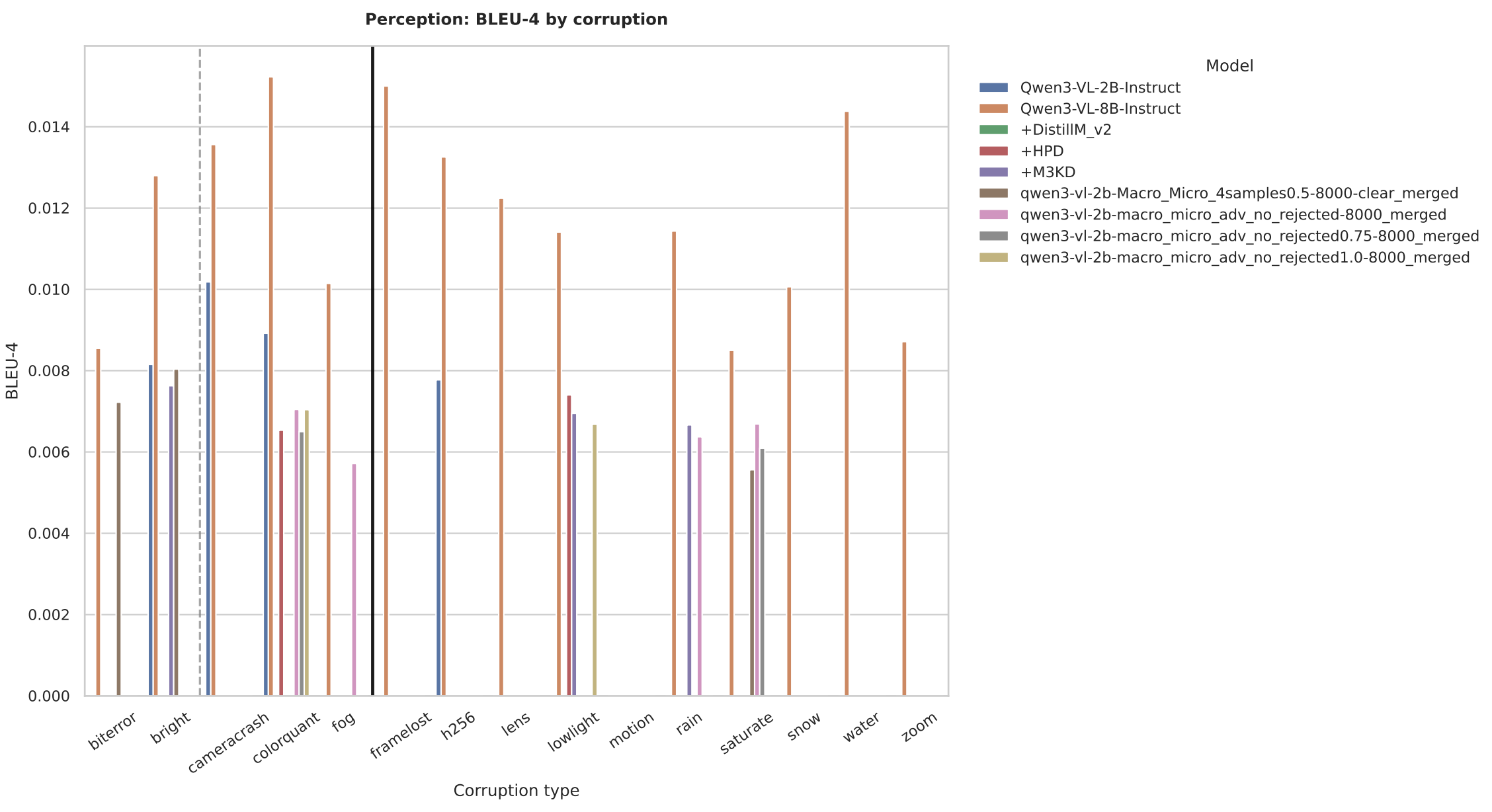
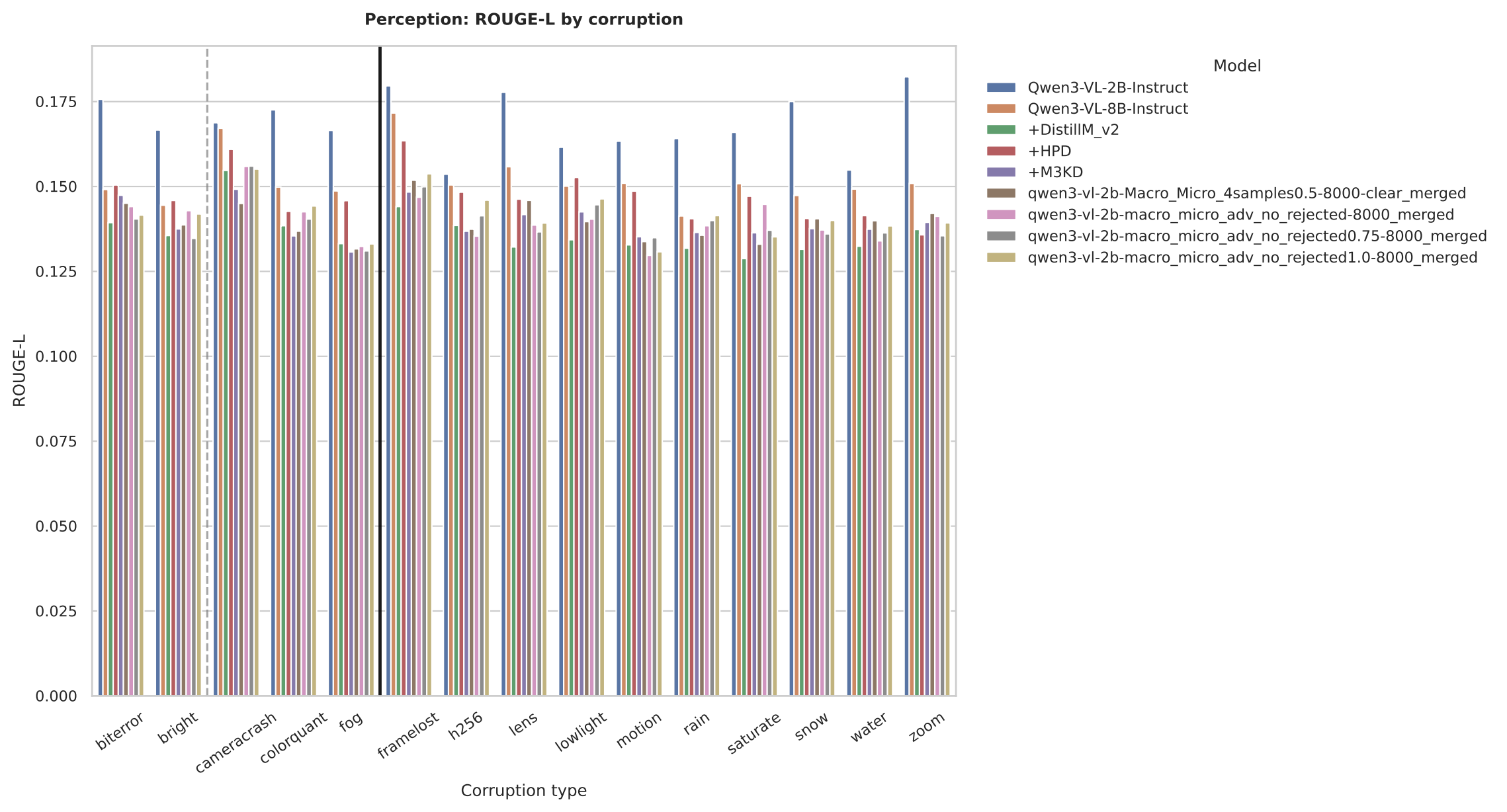
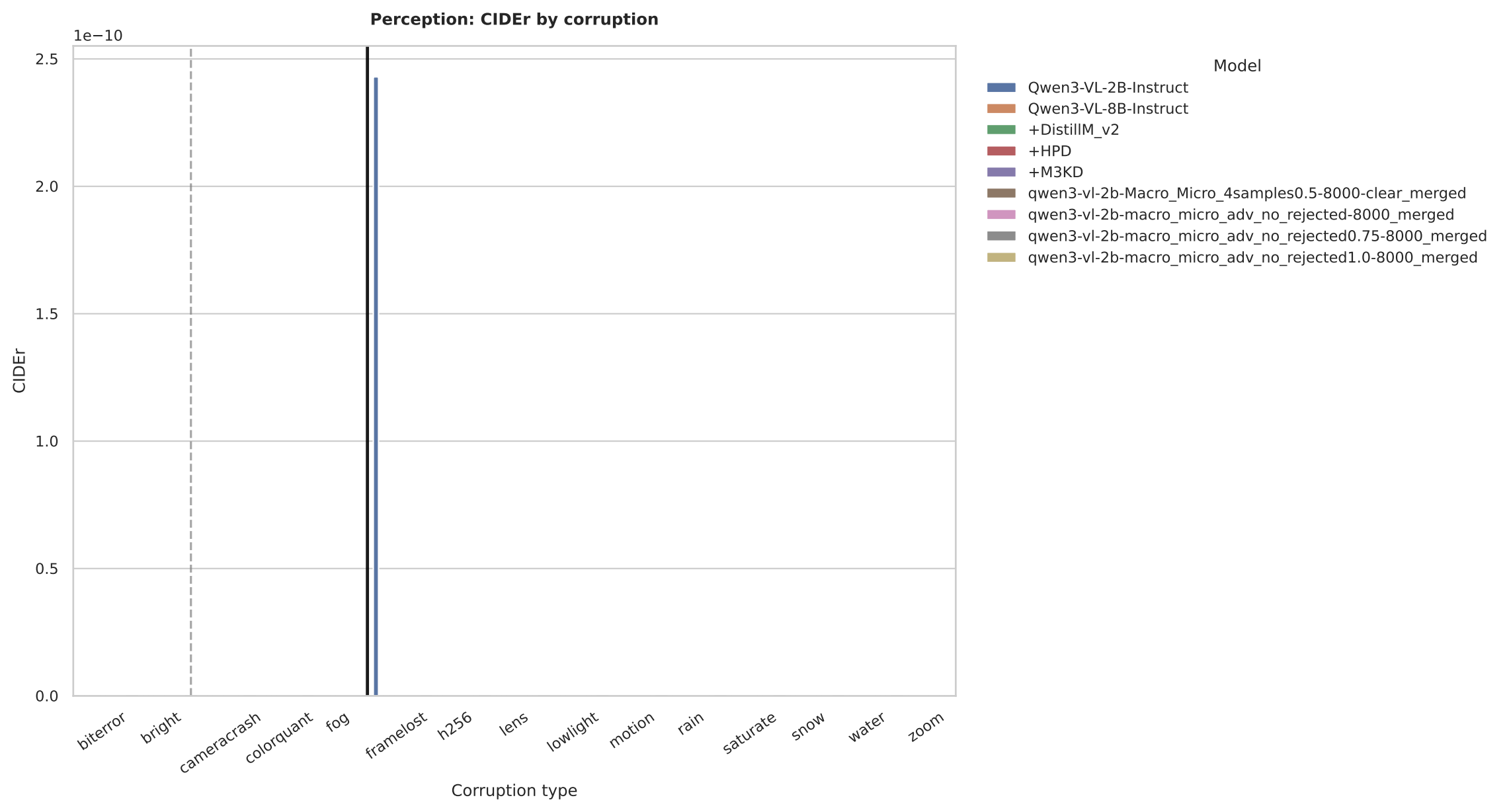
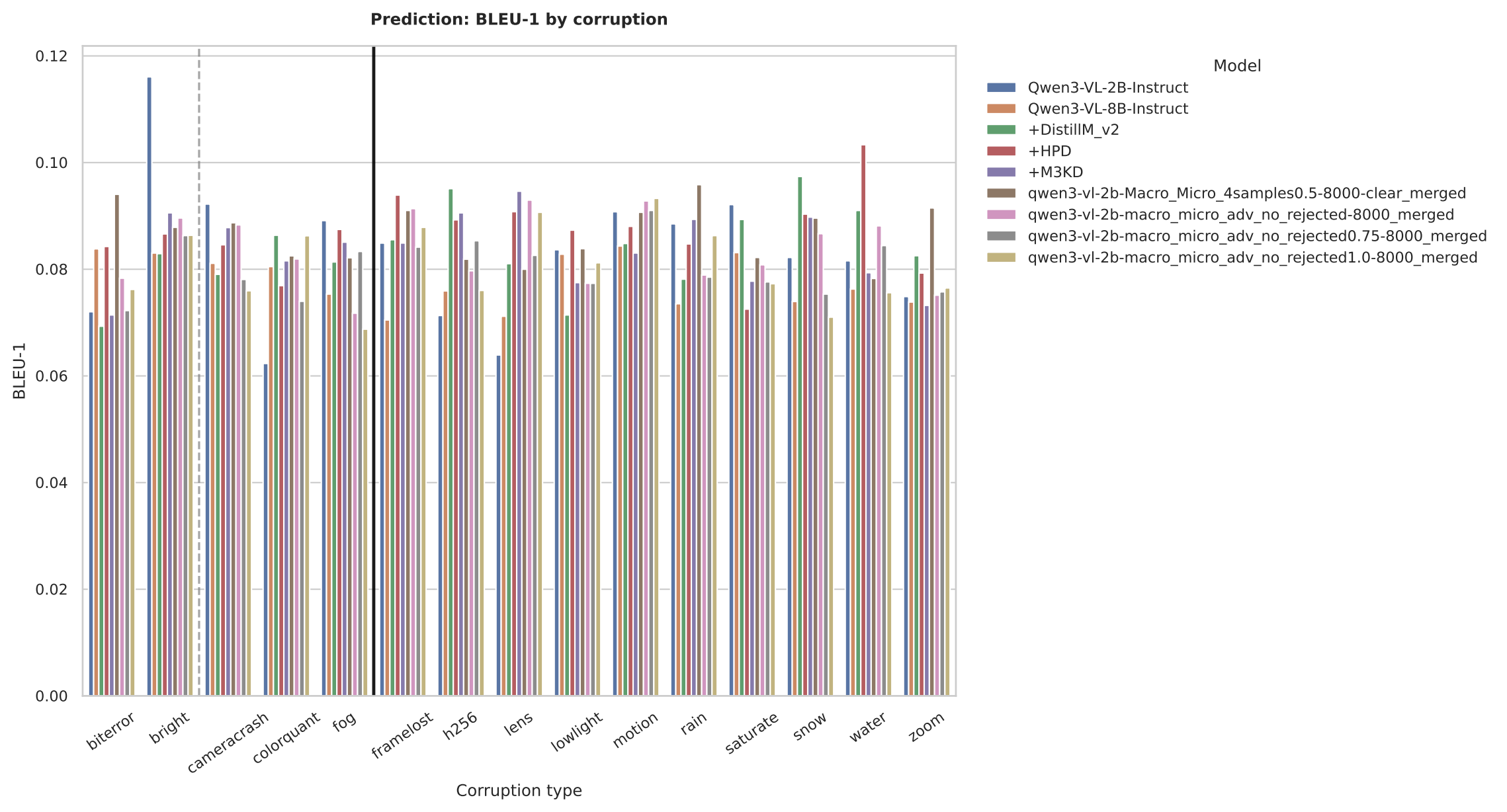

Perception: MCQ by corruption
MCQ
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged

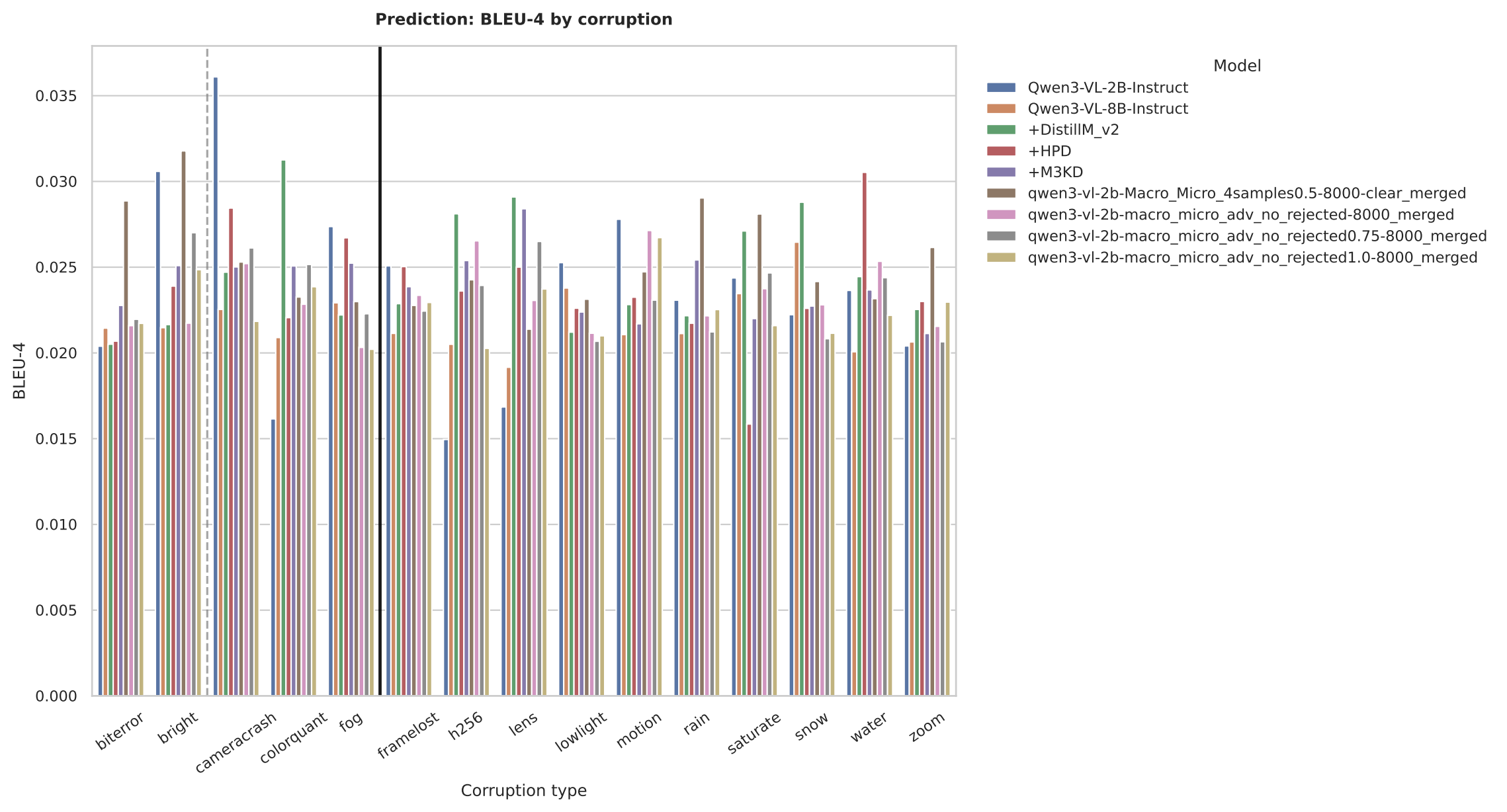

Perception: BLEU-1 by corruption
BLEU-1
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged

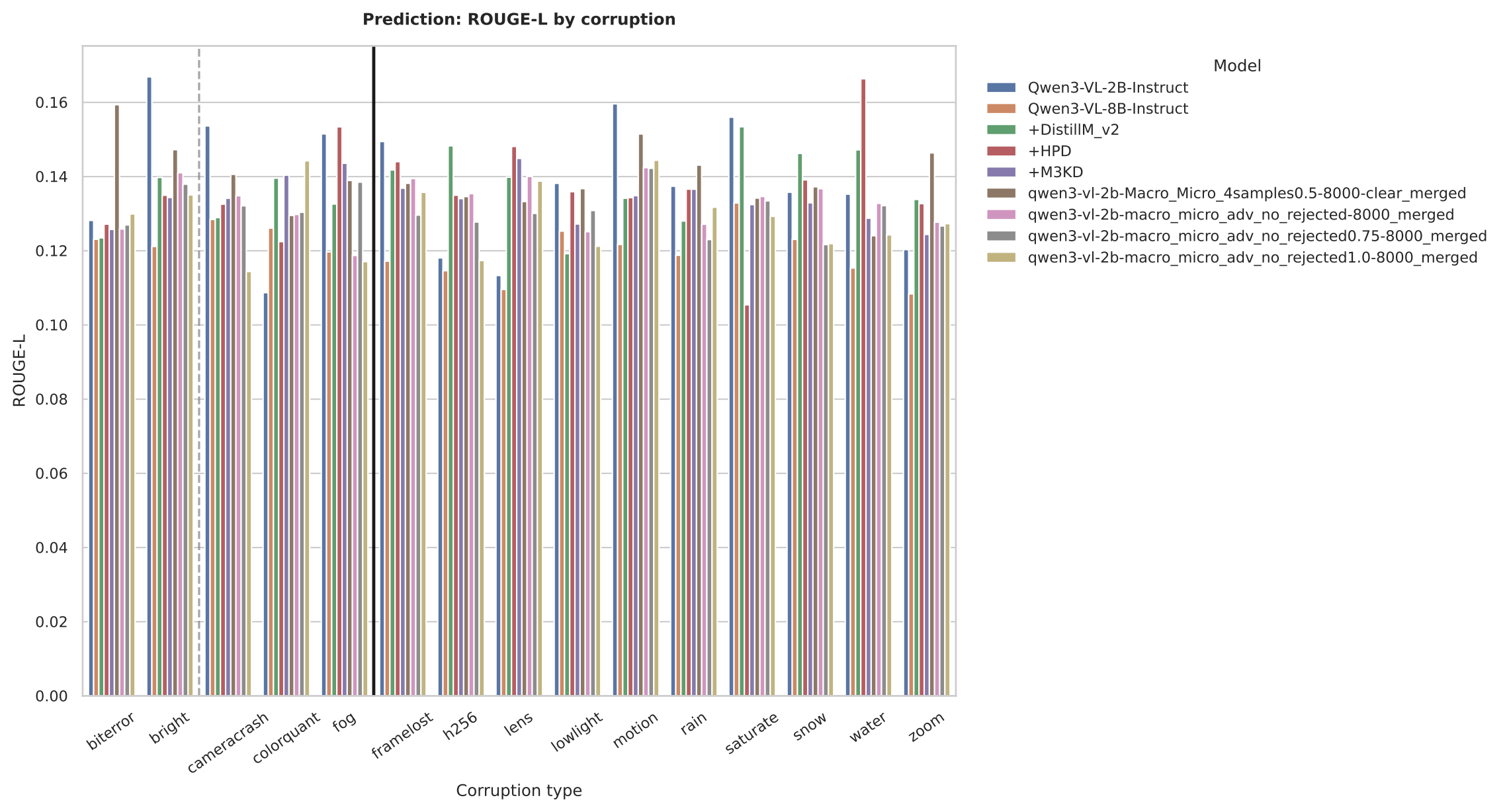

Perception: BLEU-4 by corruption
BLEU-4
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.014
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged

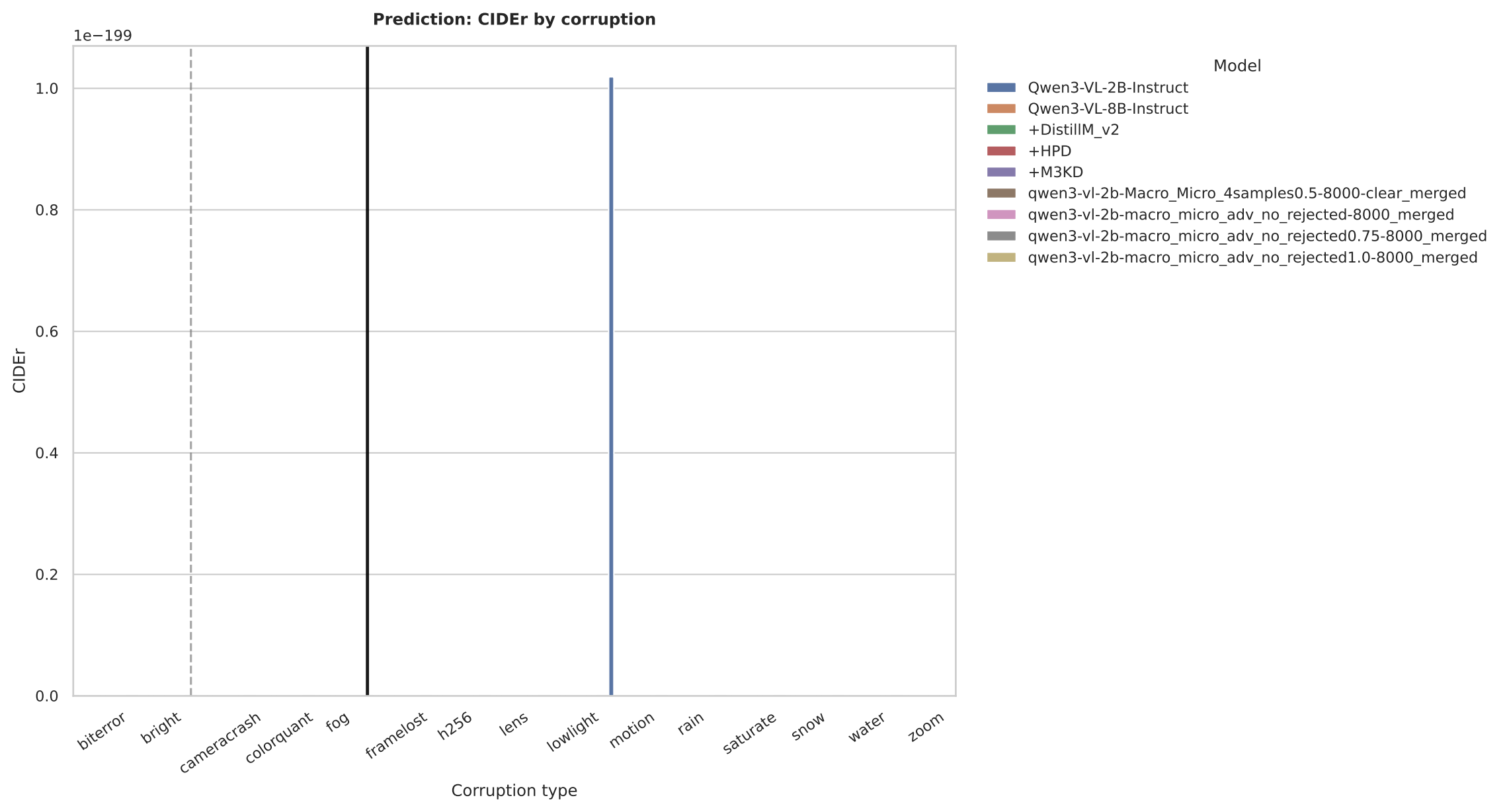

Perception: ROUGE-L by corruption
ROUGE-L
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged

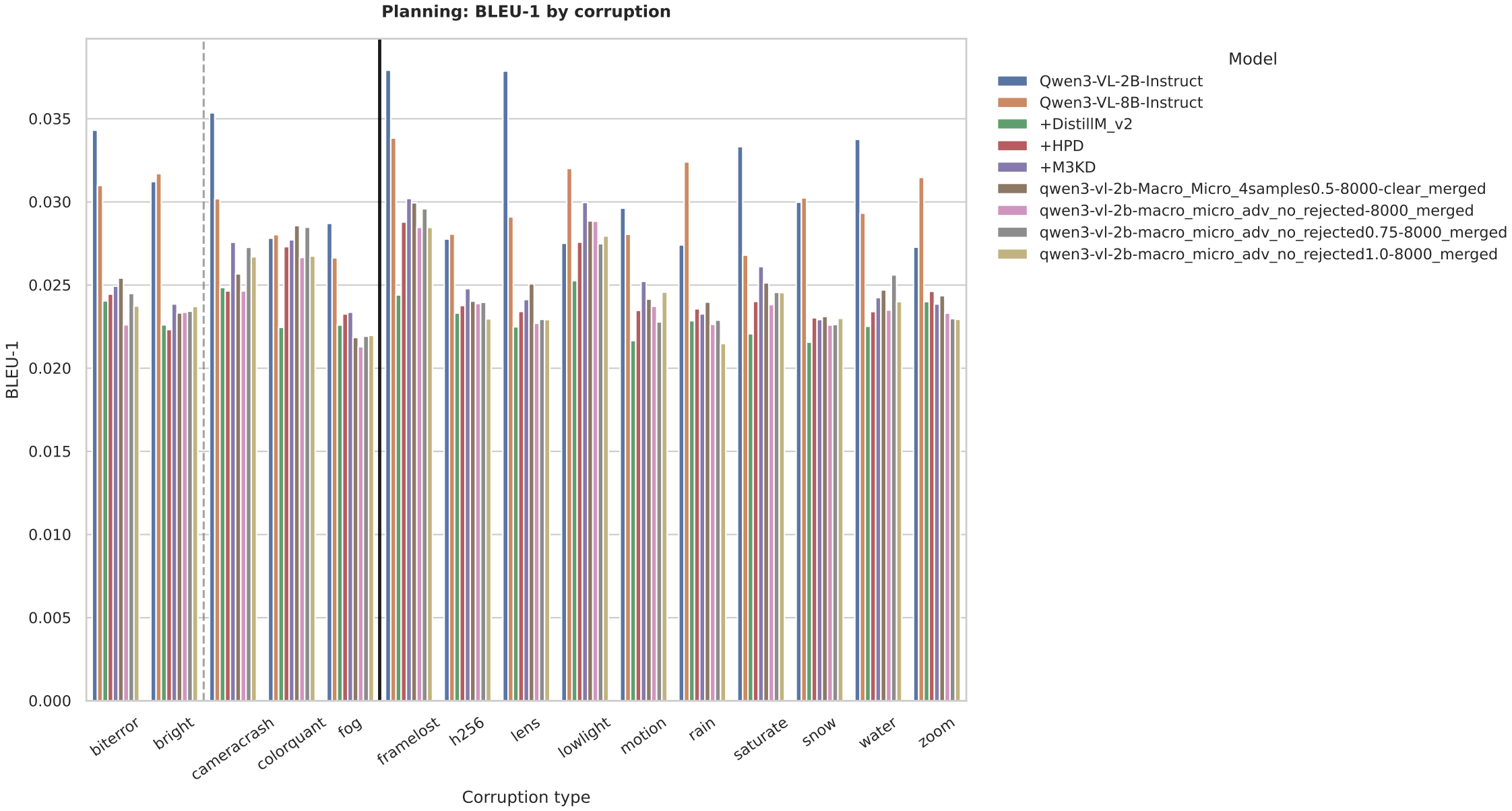

Perception: CIDEr by corruption
1e−10
2.5
2.0
1.5
1.0
0.5
0.0
CIDEr
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged

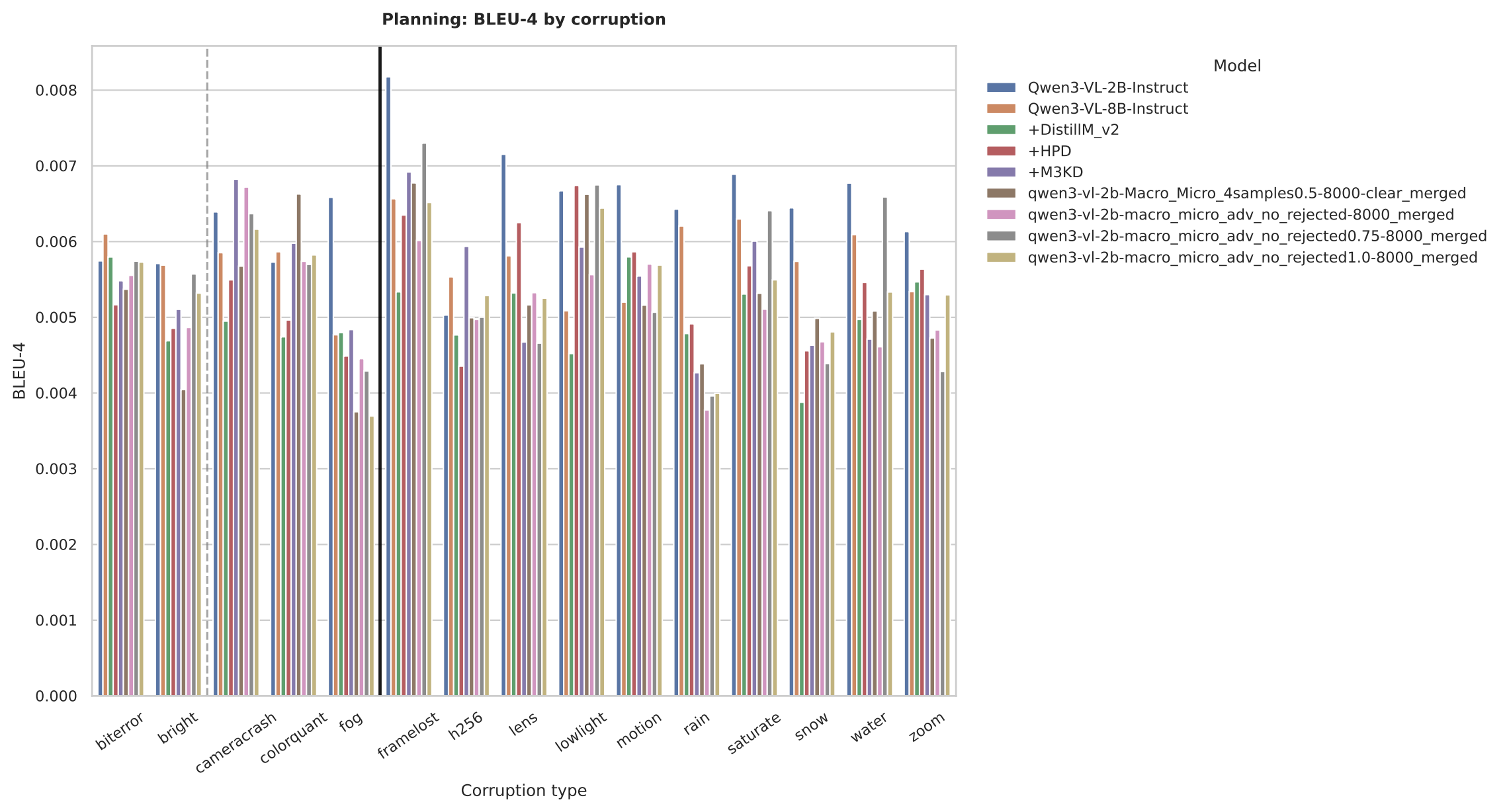

Prediction: BLEU-1 by corruption
BLEU-1
0.00
0.02
0.04
0.06
0.08
0.10
0.12
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged

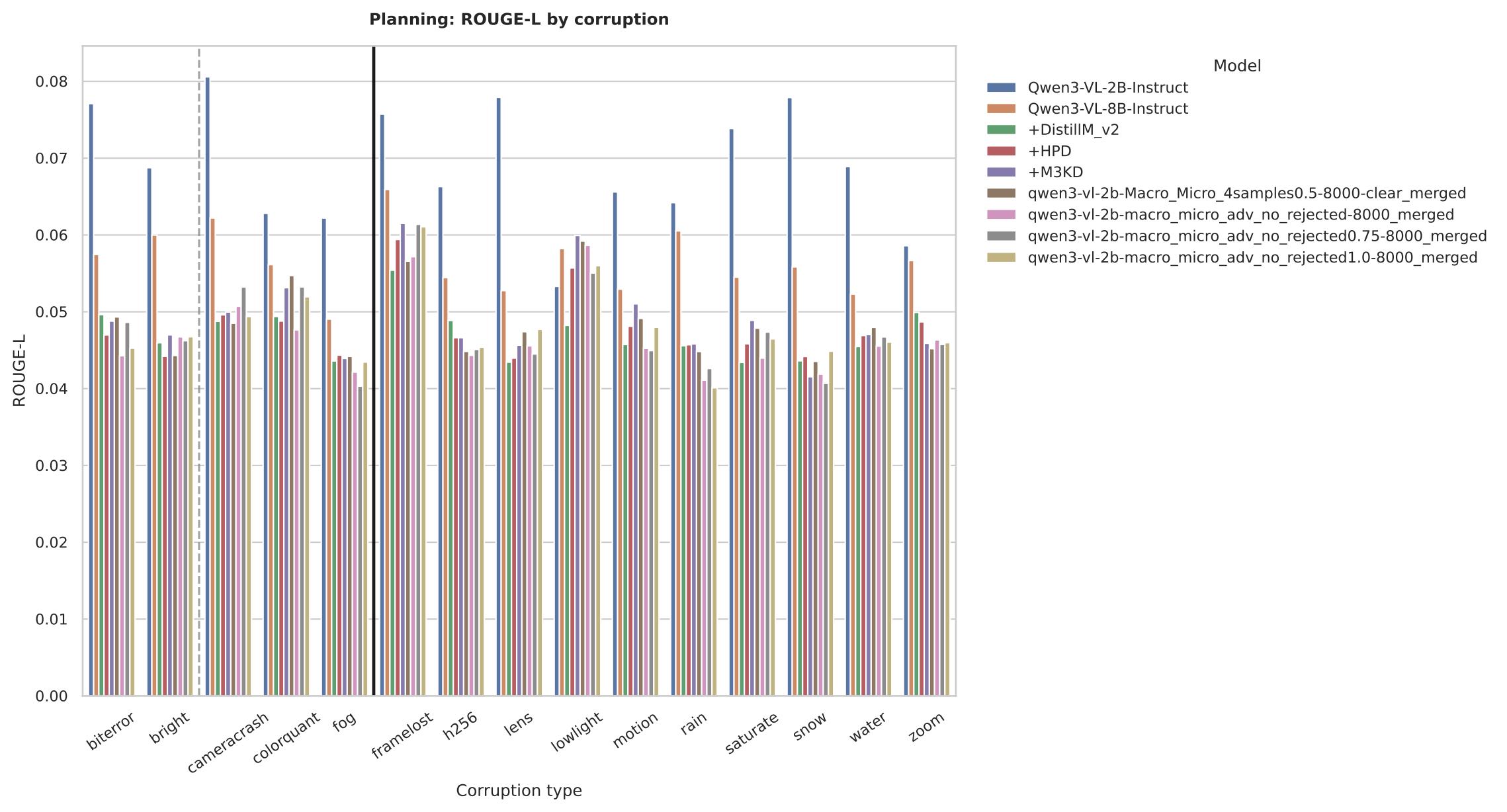

Prediction: BLEU-4 by corruption
BLEU-4
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.035
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged

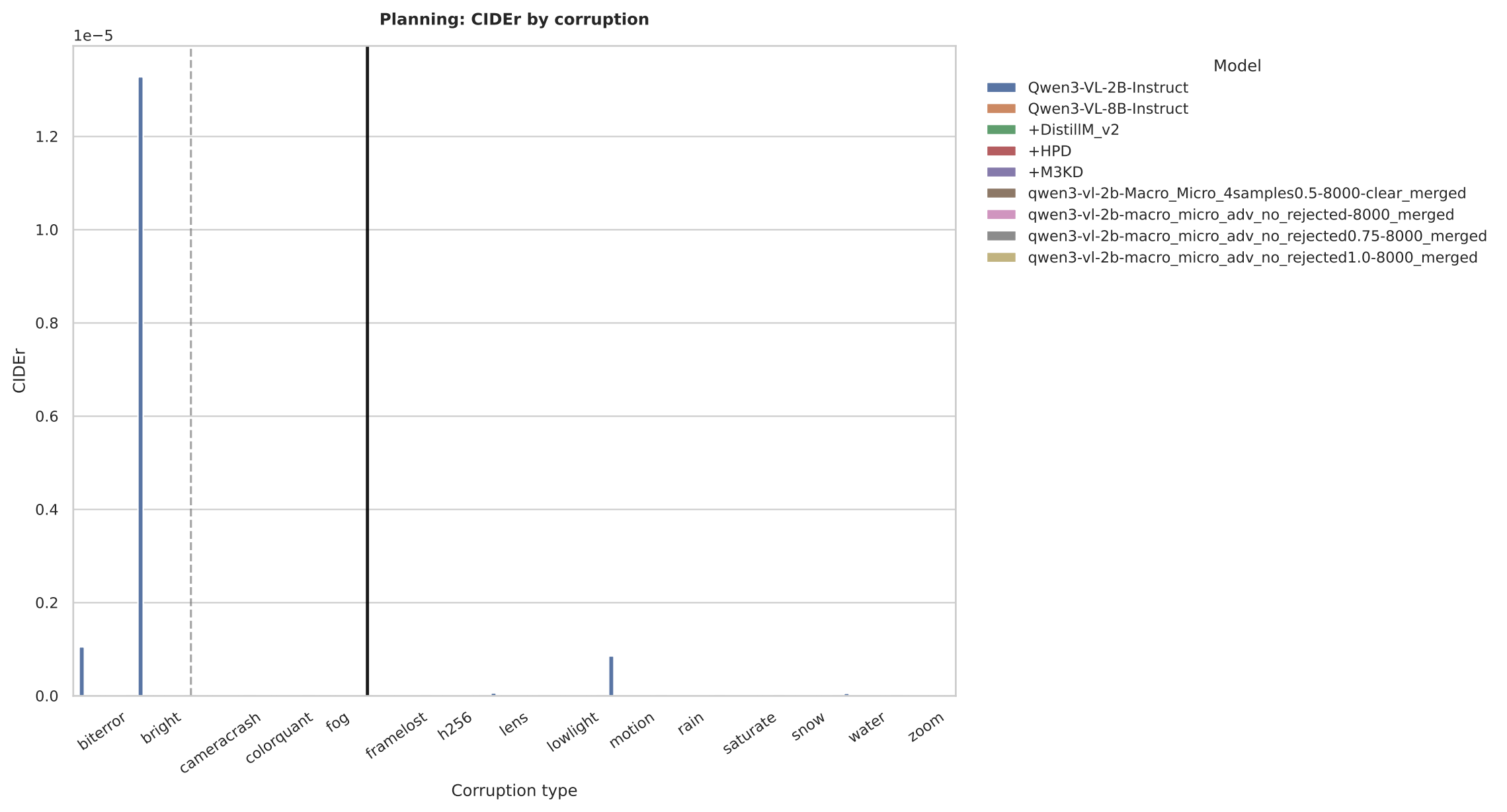

Prediction: ROUGE-L by corruption
ROUGE-L
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged

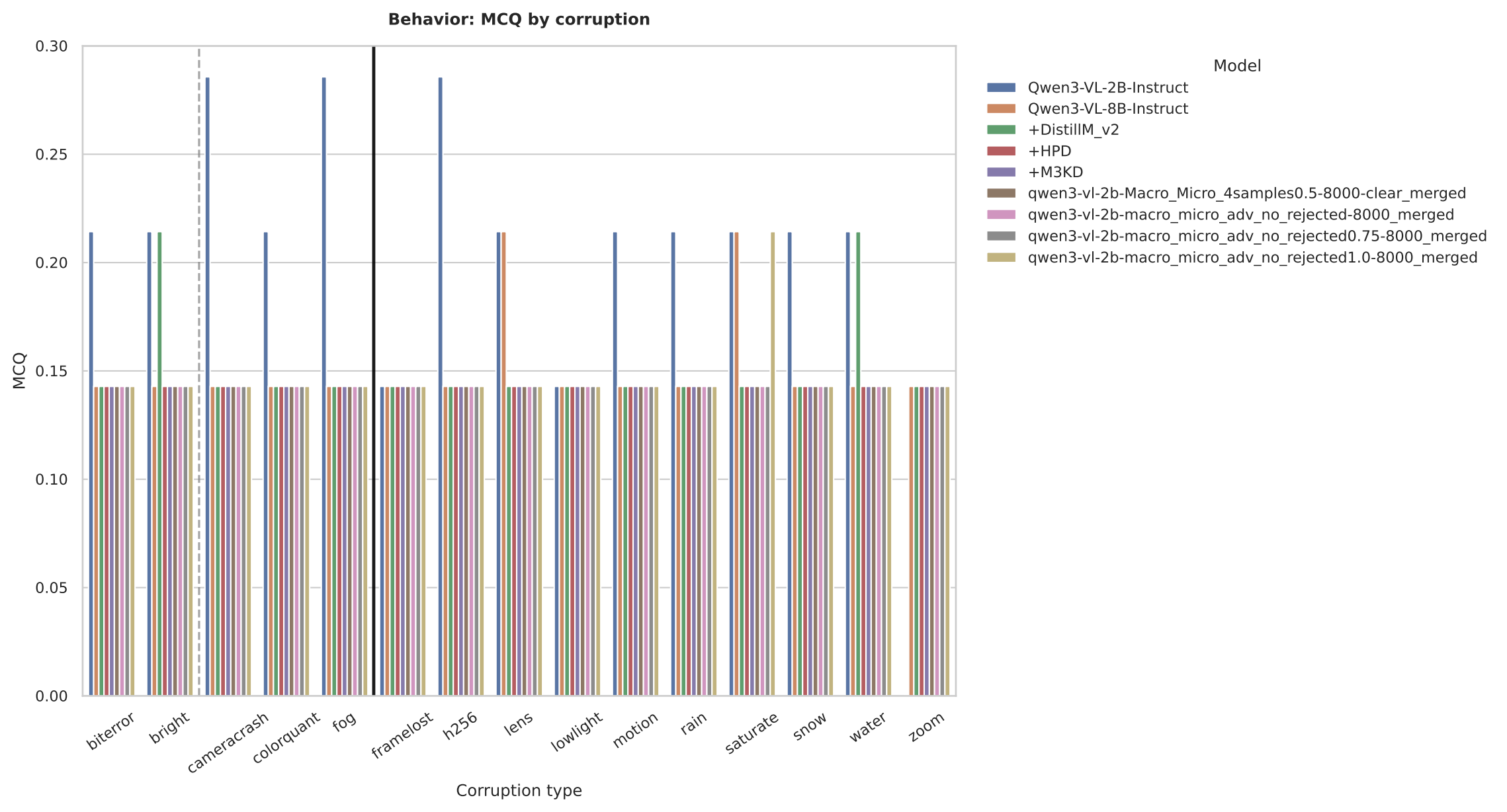

Prediction: CIDEr by corruption
1e−199
1.0
0.8
0.6
0.4
0.2
0.0
CIDEr
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged


Planning: BLEU-1 by corruption
BLEU-1
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.035
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged


Planning: BLEU-4 by corruption
BLEU-4
0.000
0.001
0.002
0.003
0.004
0.005
0.006
0.007
0.008
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged


Planning: ROUGE-L by corruption
ROUGE-L
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged


Planning: CIDEr by corruption
1e−5
CIDEr
0.0
0.2
0.4
0.6
0.8
1.0
1.2
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged


Behavior: MCQ by corruption
MCQ
0.30
0.25
0.20
0.15
0.10
0.05
0.00
biterror
bright
cameracrash
colorquant
fog
framelost
h256
lens
lowlight
motion
rain
saturate
snow
water
zoom
Corruption type
Model
Qwen3-VL-2B-Instruct
Qwen3-VL-8B-Instruct
+DistillM_v2
+HPD
+M3KD
qwen3-vl-2b-Macro_Micro_4samples0.5-8000-clear_merged
qwen3-vl-2b-macro_micro_adv_no_rejected-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected0.75-8000_merged
qwen3-vl-2b-macro_micro_adv_no_rejected1.0-8000_merged